Results

The error analysis covers the 30 public law and civil law tasks of the de-core-45 study with eight arms per task (Table 1). Every content criterion that the judge committee rated FAIL was assigned to an error class independently by two AI annotators; an assigned FAIL is called a finding. Most tables count findings, not errors: the same gap can be captured by several criteria of one answer. Table 6 counts each such gap once. Findings on which the annotators disagreed were decided by the author, so every finding enters the frequencies.

Main findings

- Three out of four findings concern missing content: K1 accounts for 75% of the counted findings. Misstated rules (K3, 8%), misapplied facts (K2, 8%) and citation errors (K5, 6%) follow far behind; K4 and K6 together make up 0.7%.
- Within K1, in 57% of the findings with a level the subject matter of the required examination is not named at all (K1a); in 26% the standard is missing, in 15% the application, and only in 1% the conclusion alone.
- Counting each gap in an answer once leaves 2,896 errors from 3,330 findings with an error class (Table 6). The merging concerns almost only K1: its 2,578 findings make 2,154 errors.
- The distribution is similar across the eight arms (Table 5). The K3 share ranges from 3% (Sol baseline) to 10% (DeepSeek baseline).
- The two annotators agree on the class of 94% of the findings (α 0.850), with the K1 levels included on 80%. Agreement is lower in public law than in civil law (Table 2).
- 3% of the counted findings are FAILs that the answer text does not confirm (OK), a finding about the automatic rating.

Scope and agreement

Table 1 gives the scope of the analysis, Table 2 the agreement of the two annotators by area of law.

Table 1: Scope: tasks, answers and content ratings of the eight arms.

Area of law	Tasks	Answers	Content ratings	Content FAILs	FAIL share
Public law	15	119	4,370	1,447	33.1%
Civil law	15	120	4,552	1,997	43.9%
Total	30	239	8,922	3,444	38.6%

Note. Eight arms per task; one arm has no submission for one task. A content rating is one content criterion of one answer rated PASS or FAIL by the judge committee. Every content FAIL is one finding of the annotation.

Table 2: Agreement of the two annotators on the class of each finding.

	n	Class level		With K1 levels	
		Agreement	α	Agreement	α
Public law	1,447	92.1%	0.830	78.0%	0.727
Civil law	1,997	94.8%	0.867	81.3%	0.734
Total	3,444	93.7%	0.850	79.9%	0.735

Note. Percent agreement and Krippendorff's α (nominal). The class level merges K1a to K1d into K1.

Error distributions

Table 3 gives the class frequencies by area of law and Table 4 the four levels of K1. Table 5 gives the class shares per arm. Arms: SB Sol baseline, SA Sol agent, TB Terra baseline, TA Terra agent, DB DeepSeek baseline, DA DeepSeek agent, CN Codex without web, CW Codex with web.

From findings to errors

When a gap or a wrong statement makes several criteria of one answer fail, the annotators flag the later findings as consequences of the first ("Folgefehler von"). Table 6 counts each error once, at its origin.

Errors hidden behind missing content

The order of checks records missing content before a misstated rule or an altered fact at the same place. Table 7 counts how often such an error stands behind a K1 finding.

Table 3: Class frequencies of the findings by area of law.

Class	Public law	Civil law	Total	Share
K1: content missing	1,036	1,542	2,578	74.9%
K2: facts misapplied	134	130	264	7.7%
K3: rule misstated	83	179	262	7.6%
K4: facts altered	2	1	3	0.1%
K5: citation missing or wrong	111	90	201	5.8%
K6: designation missing or imprecise	15	7	22	0.6%
OK: FAIL not confirmed	66	48	114	3.3%
Total	**1,447**	**1,997**	**3,444**	**100.0%**

Note. Counted are 3,444 of 3,444 findings: 2,748 on which both annotators assign the same class and 696 decided by the author. OK is not an error class: the FAIL is not confirmed by the answer text. Definitions in error-classes.pdf.

Table 4: The four levels of K1: the first missing element of the required examination.

Level	What is missing first	n	Share
K1a	subject matter not named	1,479	57.4%
K1b	standard missing	680	26.4%
K1c	application missing	386	15.0%
K1d	conclusion missing	33	1.3%
K1 with level		2,578	100.0%

Note. Of the 2,578 K1 findings, 2,578 have a level: 2,053 on which both annotators agree and 525 decided by the author. The levels are separated less reliably than the classes (Table 2).

Table 5: Content FAILs and class shares by arm.

Arm	Content FAILs	Share of the arm's counted findings					
		K1	K2	K3	K5	K4, K6	OK
Sol baseline	368	76.1%	8.2%	3.3%	7.6%	1.1%	3.8%
Sol agent	316	75.3%	6.0%	7.6%	5.1%	0.9%	5.1%
Terra baseline	466	76.6%	6.0%	7.3%	7.1%	0.4%	2.6%
Terra agent	483	72.3%	10.6%	8.3%	5.8%	0.6%	2.5%
DeepSeek baseline	620	76.5%	5.6%	10.5%	4.5%	0.5%	2.4%
DeepSeek agent	527	68.9%	10.2%	9.7%	5.7%	0.9%	4.6%
Codex without web	373	79.4%	5.1%	6.7%	5.6%	0.8%	2.4%
Codex with web	291	75.9%	9.6%	3.8%	5.8%	0.7%	4.1%

Note. Content FAILs: all content FAILs of the arm over its submitted answers (30 per arm, DeepSeek agent 29). Shares refer to the arm's counted findings (unanimous or decided) and add up to 100 per row. The shares show which kinds of errors an arm makes, not how good it is.

Table 6: Findings and errors: the same gap in one answer counted once.

Class	Findings	Errors
K1: content missing	2,578	2,154
K2: facts misapplied	264	255
K3: rule misstated	262	262
K4: facts altered	3	3
K5: citation missing or wrong	201	200
K6: designation missing or imprecise	22	22
Public law	1,381	1,321
Civil law	1,949	1,575
Total	**3,330**	**2,896**

Note. Findings with an error class (K1 to K6). A finding flagged as a consequence of another error finding of the same answer (flag "Folgefehler von") does not count again as an error; only its origin counts, with the origin's class. Where only one annotator set the flag (147 findings), the author decided by the rule of the error classes. Findings and errors are different units and are never added.

Table 7: Misstated rules (K3) and altered facts (K4) hidden behind missing content (K1) at the same place.

Level	K1 findings	Hit, both annotators	Hit, at least one	Share
K1a	1,479	25	64	1.7% to 4.3%
K1b	680	3	47	0.4% to 6.9%
K1c	386	7	29	1.8% to 7.5%
K1d	33	2	6	6.1% to 18.2%
K1	2,578	37	146	1.4% to 5.7%

Note. The order of checks records only the primary error. For every finding they assigned to K1, the annotators additionally answered whether a K3 or K4 is present at the same place; the answer does not change the class. Counted over all K1 findings, including those decided by the author; for 49 of them only one annotator assigned K1 and answered. Of the 146 findings with a hit by at least one annotator, 114 involve a misstated rule and 36 an altered fact.